Used Car Price Intelligence & Prediction System

Statistical Testing Report

Project Phase: Phase 1
Dataset: train.csv

Prepared as a record of the statistical hypothesis test implemented in the project analysis.

1. Introduction

This report documents the hypothesis test included in the Phase 1 analysis of the Used Car Price Intelligence & Prediction System. The test checks whether the listed prices of cars with automatic and manual transmissions differ statistically.

The main dataset used for this comparison is **train.csv**. The project script separates the non-missing selling-price values into two independent groups based on transmission type and applies the Mann–Whitney U test.

2. Objective

To compare the selling-price distributions of automatic and manual used cars and determine whether the observed difference is statistically significant at the 5% significance level.

3. Hypotheses

Null hypothesis (H■): The selling-price distributions of automatic and manual cars are the same.

Alternative hypothesis (H■): The selling-price distributions of automatic and manual cars are different.

This is a two-sided test because the analysis checks for a difference in either direction, rather than assuming in advance that one transmission type has higher prices.

4. Statistical Test Used

Mann–Whitney U test (two-sided)

The Mann–Whitney U test is a non-parametric test used to compare two independent groups. It works with the ranks of observations rather than requiring the data to follow a normal distribution. It is suitable for comparing two independent price groups when the price data may be skewed.

The test compares the overall distributions. A difference in the test result should not automatically be described as a difference in means. Group medians are also reported to help describe the typical price in each group.

5. Variables Used

Item	Description
Grouping variable	Transmission
Group 1	Automatic cars
Group 2	Manual cars
Comparison variable	Selling price (Price)

6. Methodology

The analysis follows these steps:

1. Load the main dataset and use the cleaned selling-price values available to the analysis.
2. Separate the non-missing prices into automatic and manual transmission groups.
3. Record the number of observations in each group.
4. Calculate the median selling price for each group.
5. Apply the two-sided Mann–Whitney U test to the two groups.
6. Record the U statistic and p-value.
7. Compare the p-value with the significance level of 0.05.

7. Significance Level and Decision Rule

The significance level is set to α **= 0.05**. This is the threshold used by the project script.

If p-value is less than 0.05: Reject the null hypothesis. The result is statistically significant, indicating evidence that the two price distributions differ.

If p-value is greater than or equal to 0.05: Do not reject the null hypothesis. The analysis does not provide sufficient evidence of a difference between the two price distributions.

A p-value is not the probability that the null hypothesis is true. It measures how unusual the observed result would be if the null hypothesis were true.

8. Results

The project script is designed to report the following values. The actual output values were not present in the report/source material available when this document was prepared, so they are left for insertion from the program output rather than being estimated or invented.

Result	**Value from program output**
Number of automatic cars	To be filled from output
Number of manual cars	To be filled from output
Median price – automatic	To be filled from output
Median price – manual	To be filled from output
Mann–Whitney U statistic	To be filled from output
p-value	To be filled from output
Decision at α = 0.05	To be determined from p-value

9. Interpretation

The final interpretation must be based on the p-value generated by the actual program run. If the p-value is below 0.05, the report can state that the price distributions differ significantly. If it is 0.05 or above, the report should state that the test did not find sufficient statistical evidence of a difference.

The group medians should be included when explaining the direction of the observed difference. The U statistic and p-value alone do not explain how large or practically important the difference is.

10. Limitations

• The test compares the two transmission groups without controlling for other factors such as car age, brand, kilometres driven, engine size or power.
• A statistically significant result does not show that transmission type causes a change in selling price.
• The test compares distributions; it should not be interpreted as a direct test of the difference between group means.
• The reliability of the comparison depends on correct transmission labels and valid price values.
• Statistical significance does not necessarily mean that the price difference is large or important in practice.

11. Conclusion

The Phase 1 project uses a two-sided Mann–Whitney U test to compare selling-price distributions between automatic and manual used cars. The test is appropriate for a comparison of two independent groups and does not require the price data to be normally distributed. The final conclusion will be completed after the group counts, medians, U statistic and p-value are taken from the actual program output.

The result provides one statistical check within the broader exploratory analysis. It does not establish a causal relationship and does not measure the predictive accuracy of a machine-learning model.

12. Software Used

Python, Pandas and SciPy are used in the project analysis. The Mann–Whitney U test is performed through SciPy's statistical functions.